# A MAP OF THE RANGE, NOT A WAY UP THE WALL: SIX WEEKS OF AI AGENTS ON 373 OPEN PROBLEMS IN ANALYSIS

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Abstract. For six weeks, one mathematician with a master's degree directed AI coding agents at every open problem that could be found in one area of analysis: Hardy, Bergman and Dirichlet-series spaces, Hankel and Helson operators, and Nehari-type theorems. The agents collected 373 problems, found that 64 of them had already been settled by others, and produced answers to 141 more. We classify these answers by how they were obtained and by how much they matter. Most are easy in hindsight: short arguments from known tools, results assembled from the literature, or counterexamples that are easy to check once found. Six are, in our judgement, of real significance. On the hardest problems, days of work by several agents produced maps of what fails, not proofs. We conclude that current agents are very good at mapping a field and clearing its long tail, and that a hard problem stays hard. All results of ours are sketches until they appear on arXiv or in a journal; this is a living document, updated as the people who asked the questions respond.

## 1. Introduction

Terence Tao has described research mathematics as a range of mountains:

> Tao gave the analogy of mathematicians trying to climb "a big mountain range with lots of tall mountains and lots of foothills". Humans can only climb one step at a time, but they can plan a route to the top of a mountain like Everest. Meanwhile, Tao said, current AIs are like jumping robots. They can sometimes "parkour their way to the top of a 6-foot wall" that a human couldn't climb. But they can't do long-term strategic planning. Those 6 feet might become 10 feet, or 100, Tao imagines, but "the little jumping robots are nowhere near the Mount Everests of math" [21].

This paper is a field report on what such robots do when pointed at an entire range. Over six weeks in autumn 2026 we used Claude Code and Codex agents to find, sort and attack every open problem we could locate in one research area. The agents found 373 problems in papers, problem lists and monographs. They established that 64 were already solved, and they produced answers to 141 others.

The answers need careful reading. Our main finding is that the gain was in cartography, not in climbing:

- The agents were very good at mapping a field: finding problems, classifying them, checking the literature, and clearing problems that were easy but had received little attention.
- Most of their answers are low-hanging fruit. Of the 141, we classify 6 as significant, 62 as useful, and 73 as minor.
- On problems where experts had made partial progress and stopped, the agents produced reformulations, special cases and catalogues of obstructions, not proofs.

Breadth explains the counts. Throw a hundred balls at a hundred baskets and some go in; that is useful, and different from getting better at the shot that matters. Tao's description of current tools as "artificial general cleverness" [23] fits what we saw.

We therefore present this method as the first step of a research project, not the last: a fast way to get to know a field and to find questions worth working on. Section 3 describes how

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the problems were found and sorted, Section 4 the workflow, Section 5 the results and their classification, Section 6 the hard problems, and Section 7 verification and credit. Section 8 places the work among other reports, and Section 9 lists its limits.

## 2. Setting

The area is the theory of Hardy and Bergman spaces on the disc and on the infinite polydisc, Dirichlet series viewed as functions on the infinite torus, and the Hankel, Helson and composition operators acting on them. A typical question asks for the sharp constant in an inequality, whether an extremal function can have zeros, or whether a bounded bilinear form has a bounded symbol. The active literature is a few hundred papers, much of it from a small number of research groups.

One person directed the work. He chose targets, judged whether a result mattered, steered agents that were stuck, read everything before it went out, and handled all contact with other mathematicians. The agents were Claude Code (Claude Opus 5.5 as the main session, with smaller Claude models for cheap subtasks) and Codex (several GPT models), often running at the same time in the same repository. The record of the work is the version history, with about 1000 commits, together with the logs: 14 Claude Code sessions with 393 subagents, and 586 Codex sessions, 420 of them subagents.

## 3. Building the map

The problems came from four sources, in this order.

(1) *Stated problems in a local paper collection.* Agents searched each paper for environments labelled Conjecture, Question, Problem or Open problem, and for phrases such as "remains open" and "it would be interesting to". Each candidate was recorded with the paper, the exact location, and a quote.
(2) *The papers those cite*, with the same extraction.
(3) *Synthesis passes.* Agents read the recent papers of the core authors "between the lines". This produced *implicit* questions, which a remark suggests without posing, and *inferences*: paper A proves one statement and paper B another, so a third should follow.
(4) *Problem lists and monographs*: workshop problem sessions, a classical problem book, and the open-problem sections of two monographs.

Each problem became one Markdown file with validated metadata. The metadata records:

- the theme, the kind (298 stated, 55 implicit, 20 inferences) and the origin (289 asked by others, 84 by us);
- tags, an impact score and a difficulty score;
- a dated literature check with quoted evidence;
- the stage reached in our own work;
- once anyone settled the problem, how far the resolution is trusted: journal, arXiv, or our own.

Moving from a single catalogue file to one file per problem let several agents and two different tools edit the database at once.

## 4. The workflow

Each problem was worked in sessions. A session begins by reading the record, the source passage and the log of earlier attempts, and by writing a three-line plan. It ends by rewriting a short *state of play* and appending one line to the *attempt log*: the route, the outcome, the reason, and the next step. A claimed result had to become a *solution sheet*: the problem as posed, the answer as a theorem, a complete proof with exact citations, and nothing else, built through a lint-gated LaTeX pipeline. A fresh agent with a fixed brief then refereed it.

The referee re-read the question in its source, re-derived each step, reran any certificate, and decided whether the answer was full or partial.

On hard problems several agents worked in parallel *lanes*, each on one approach, with a coordinating session redirecting them. When lanes drifted into cataloguing obstructions, a *prove-mode* brief told each to commit to one strategy and push it to a theorem, or to a proof complete except for one precisely stated lemma. The full instructions are published with this paper as a reusable set of agent playbooks.

## 5. Results

5.1. **What the literature check found.** The difficulty assigned to a problem before anyone attempted it predicted the outcome well (Table 1). It also showed why easy problems are easy: nine of the twelve problems first rated easiest already had a published answer. Tao observed the same correlation on the Erdős problems: the problems current tools can solve are "precisely the category of nominally open problems that are most likely to have been solved in the literature" [25]. The literature check also caught one problem recorded as resolved by a paper that had since been withdrawn.

| Initial difficulty | Settled by others | Settled by us | Partial | Open |
|---|---|---|---|---|
| 1 | 9 | 3 | 0 | 0 |
| 2 | 4 | 15 | 2 | 0 |
| 3 | 7 | 27 | 19 | 7 |
| 4 | 4 | 29 | 75 | 6 |
| 5 | 0 | 3 | 15 | 3 |

Table 1. Outcome by the difficulty assigned before any attempt, for the 275 records that had one.

5.2. **What kind of results were these?** We classified each of the 141 answers along three axes.

- *Sign*: 82 proofs, 42 counterexamples, and 17 determinations of a value, set or order.
- *Kind*, meaning how the answer was obtained:
  - 20 assembled from the literature;
  - 58 routine new arguments;
  - 44 explicit constructions or certificates, which are the work to find and easy to check;
  - 19 substantial new arguments.
- *Significance*:
  - 6 high: paper-level answers to someone else's question;
  - 62 medium: full answers worth a short note;
  - 73 low: our own questions, conditional or elementary answers, and corollaries.

The classification is ours and preliminary; Table 2 gives the cross-tabulations.

Three things stand out.

- *Counterexamples are the natural strength.* Thirty-nine of the 42 counterexamples are explicit constructions or certificates. Searching a space of candidates and checking each one is exactly what agents do well, and what earlier work on machine-assisted discovery exploited [1, 13, 27].
- *Few proofs are substantial.* Only 17 of the 82 proofs needed a genuinely new argument; the rest are routine or assembled from the literature.
- *The significant results needed direction.* Of the six results we rate high, four are proofs and two are counterexamples. Three of the six came from focused sessions in which

| | Literature | Routine | Construction | Substantial | Total |
|---|---|---|---|---|---|
| Proof | 17 | 43 | 5 | 17 | 82 |
| Counterexample | 0 | 1 | 39 | 2 | 42 |
| Determination | 3 | 14 | 0 | 0 | 17 |

| | High | Medium | Low | Total |
|---|---|---|---|---|
| Proof | 4 | 33 | 45 | 82 |
| Counterexample | 2 | 24 | 16 | 42 |
| Determination | 0 | 5 | 12 | 17 |
| Asked by others | 6 | 57 | 26 | 89 |
| Asked by us | 0 | 5 | 47 | 52 |

TABLE 2. The 141 answers by sign and kind (top) and by sign, origin and significance (bottom).

the human directed one problem with the asker's question in view, not from the bulk campaign over the database.

The six high-significance results are:

- that extremal functions for bounded linear functionals on $H^p$, $0 < p < 1$, are outer, answering a published conjecture in a stronger form, written jointly with the asker;
- a sharp bound on the number of zeros of extremals in a prescribed-coefficient problem;
- the finite-exponent limits of sharp Bernstein–Nikolskii constants;
- the exact order of growth of the $H^1$ norms of partial-sum operators for Dirichlet series;
- a counterexample to a positive-symbol Nehari question, with a computer certificate in outward-rounded arithmetic;
- that optimal polynomial approximants in $H^p$ can vanish in the disc for every $p \neq 2$.

Each has been sent to the people who asked the question, or is being prepared for them. None is yet on arXiv.

## 6. Walls

We attacked several problems on which experts have made partial progress, for days and with several agents. Two examples, described without the technical details:

- *An endpoint Schatten-class Nehari problem.* Every route the agents found reduced to one square-function estimate in $L^1$ that is a theorem in its own right. They proved many special cases, wrote down equivalent reformulations, and catalogued exactly where each route failed. They did not find a proof, and the problem was handed back to the person who asked it.
- *A contractive restriction conjecture.* The agents proved the conjecture for two natural families of functions and refuted a stronger statement that would have implied it. The general case did not move.

In both cases the output was useful as a map. It is not progress on the wall itself, and we do not present it as such. The pattern matches Tao's image: some walls of six feet were climbed; the Everests were not approached.

## 7. Verification and credit

Producing a candidate proof was cheap; checking it was the bottleneck. This matches Tao's analysis of the problem-solving pipeline, in which generation is followed by verification, exposition, acceptance and absorption into the standard theory, and in which AI threatens

"proof indigestion" [22]. Our answers were checked by agent referees, in some cases by a second model family, and in a few cases by Lean formalization or exact-arithmetic certificates. Agent referees caught real errors: a bound used outside its range, a missing hypothesis, and above all overstated scope, where an answer to a special case was reported as an answer to the question. They rarely asked whether the claim was the right one.

For this reason every result of ours is labelled a *sketch* until it appears on arXiv or in a journal, however it was checked. Before anything is posted, the people who asked the question are told; this follows the recommendations of the Leiden Declaration on attribution, disclosure and the humanity of authorship [18]. The questions belong to those who asked them. The proofs and computations were produced by AI agents and checked by the author. Following Tao's rule of thumb [22], no result should be published until a human can give a clear, expert-level talk on it.

## 8. Related work

*Machine-assisted discovery.* Machine learning has guided human intuition in knot theory and representation theory [7], and reinforcement learning has found counterexamples to conjectures in combinatorics [27]. Language models coupled with evaluators have found new constructions [1, 13], and have been applied to many problems at once [14]. Our counterexamples are of this kind: constructions that are hard to find and easy to check.

*Formal proof.* Language models for theorem proving date back at least to [17, 20]; olympiad geometry has been solved without human demonstrations [2]. The mathlib library [19] underlies much of this, and Avigad discusses the wider "formal turn" [3].

*Evaluation.* Early evaluations of language models on mathematics include [6, 11]. Benchmarks with unpublished problems [12, 16] and the First Proof project, which has experts referee AI solutions to unpublished research problems [10], measure capability under controlled conditions. Our report is neither controlled nor a benchmark; it is one practitioner's account of a whole area.

*Research practice.* Experiments with frontier models as research assistants are collected in [5]. The Erdős problems database [8] and its AI wiki [9] are the closest analogue to our problem database; the wiki warns that "absence of past progress may reflect obscurity rather than difficulty". On the goals of mathematics beyond problem solving we follow Thurston [26], Gowers [15], Tao [22, 24] and Bessis [4]. Journalism on the current state of the field includes [21], where Igor Pak notes that language models are "fantastic in finding the right references".

## 9. Limits

This is one person's experience in one area, over six weeks, without a control group. The difficulty and significance scores were assigned by the same agents and by the author, not by independent experts. Correctness has not been audited on a random sample; we intend to do so, and the counts above should be read as counts of claimed answers. The models will be superseded within months. Finally, the area has a dense literature of recent preprints, which made the literature check unusually productive.

## Tool and computational resource disclosure

The work used Claude Code from Anthropic, with Claude Opus 5.5, Sonnet and Haiku, and Codex from OpenAI, with several GPT models; Python with mpmath, SymPy and cvxpy for numerics and certificates, Lean for one formalization, and LaTeX built through a lint gate. This paper was drafted with Claude Code from the project's records and logs and revised by the author. The agent instructions, templates and tools are available in the accompanying process repository.

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